Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2023-2024

Mid-Semester Test
(EC-2 Regular)

Course No.	: S2-23_CCZG502
Course Title	: Cloud Infrastructure and Systems Software
Nature of Exam	: Closed Book
Weightage	: 30%
Duration	: 2 Hours
Date of Exam	: 16/03/2024 (FN/AN)

No. of Pages = 1
No. of Questions = 9

Note to Students:
1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. Each answer should start from a fresh page.
3. Assumptions made, if any, should be stated clearly at the beginning of your answer.
4. If required to draw any image, then paper & pen can be used, and its photo can be attached.

Q.1. In a system, parallelization is done using 8 processors. A program has been updated with 60% parallelizable code and 40% sequential. Calculate the speed-up attained. **[2 Marks]**

Q.2. Differentiate between 'Computer Architecture' & 'Computer Organization'. **[4 Marks]**

Q.3. In a direct-mapped cache, if the main memory has 4096 blocks and each block is 128 bytes, and the cache has 64 blocks, calculate the number of bits needed for the cache tag and block offset separately. **[2 Marks]**

Q.4. Compare the Direct, Associative, and Set Associative mapping techniques. **[4 Marks]**

Q.5. Define Instruction Set Architecture (ISA). Also list its components. **[2 Marks]**

Q.6. With the increasing importance of complex workloads like artificial intelligence and machine learning, do you think RISC or CISC architectures are better positioned for the future? Justify your answer. **[4 Marks]**

Q.7. Define process. Briefly describe the process state diagram. **[2 Marks]**

Q.8. A system implements a Shortest Job First (SJF) scheduling algorithm. How can this algorithm potentially lead to starvation for longer processes, and what alternative scheduling algorithms could be considered to address this issue? **[4 Marks]**

Q.9. Construct a use-case where you would prefer using Containers over Virtual Machines. Justify your decision based on performance, scalability, and maintenance aspects. **[6 Marks]**

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